Global Shark Attacks

Exploratory Data Analysis & Power BI Dashboard

Marius Tillinger (511) & Radu Gheorghe (511)

Information Visualization
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Part I

Exploratory Data Analysis

Understanding the data before we visualise it

The dataset

Global Shark Attack File (GSAF)

- One row per recorded shark–human encounter, dating from the 1700s through 2018.
- Maintained by volunteer researchers; each case is backed by a PDF source.
- 25,723 raw rows × 24 columns.

Columns at a glance

`Date`, `Year`, `Type` (Provoked/Unprovoked/...), `Country`, `Area`, `Location`, `Activity` (surfing, swimming, fishing...), `Name`, `Sex`, `Age`, `Injury` (free text), `Fatal (Y/N)`, `Time`, `Species`.

Why this dataset?

An ideal teaching dataset

- **Tells a story** that everyone can engage with – sharks, beaches, fear, science.
- **Mixed data types**: temporal, geographic, categorical, numeric, free text – perfect for showcasing many chart types.
- **Realistically messy**: trailing whitespace, encoding issues, free-text fields, missing values. Cleaning is half the lesson.
- **Long time span** (3 centuries) lets us study how *recording*, not just the phenomenon, changes over time.

Questions we want to answer

1. Where and when do attacks happen?
2. Who are the victims (age, sex, activity)?
3. How dangerous is an attack – and is it getting safer?
4. Which species are involved most often / most lethally?

Issues found in the raw CSV

Structural

- File is encoded in `Latin-1`, not UTF-8 (names with accents broke otherwise).
- ~75 % of the rows are completely empty (trailer rows from Excel).
- Column names carry trailing spaces (`'Sex '`, `'Species '`).
- Three duplicate `Case Number` columns and two unnamed columns are pure noise.

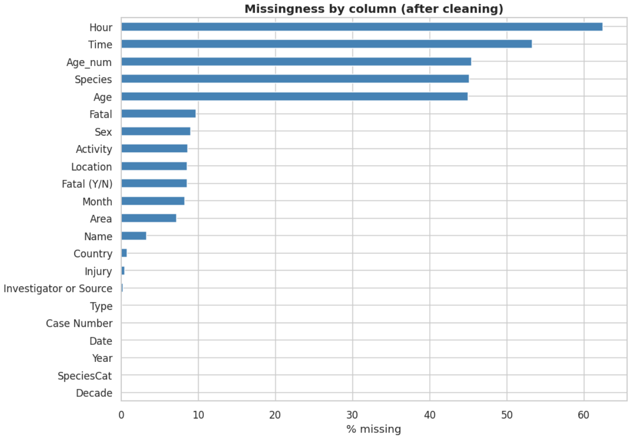
Field-level

- `Fatal (Y/N)` contains `Y`, `N`, `UNKNOWN`, `M`, `2017`, `' N'`, `y`.
- `Sex` contains `M`, `F`, `'.'`, `'lli'`, `'N'`.
- `Age` is a string: *"18 to 22"*, *"teen"*, *"middle aged"*.
- `Species` is free text; `Date` is mixed format; `Time` uses `"14h00"`.

What we did

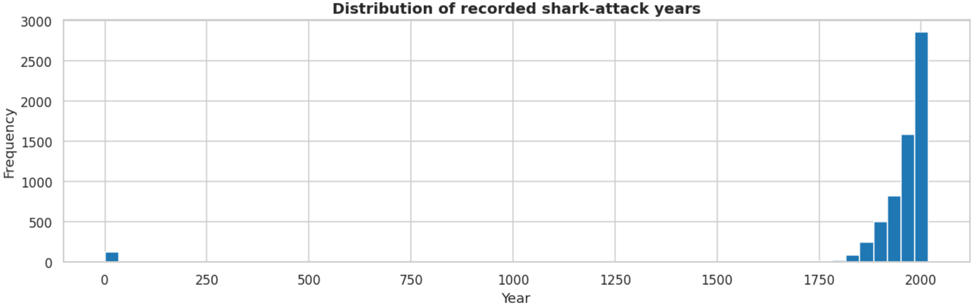
1. Stripped whitespace from all column names and string values.
2. Dropped duplicate / unnamed columns and 19,423 fully empty rows $\Rightarrow$ **6,300 usable records**.
3. Normalised `Fatal` to {Y, N, NaN} and `Sex` to {M, F, NaN}.
4. Extracted a numeric `Age_num` (regex for first digit run, 1–100).
5. Parsed `Month` from `Date` and `Hour` from `Time`.
6. Mapped free-text `Species` to a coarse `SpeciesCat` (White, Tiger, Bull, . . .).
7. Bucketed `Activity` into seven readable categories.

Missingness after cleaning



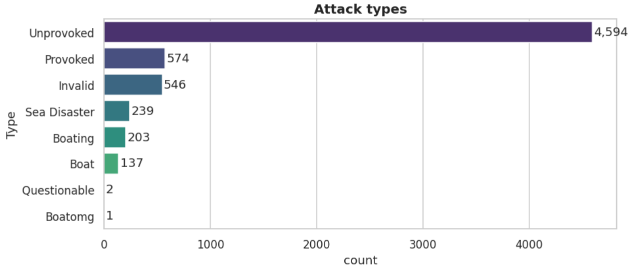
- `Time` and `Species` are the *most-missing* useful fields (∼50–55 %).
- Core analytical columns (`Year`, `Country`, `Type`, `Activity`, `Fatal`) are well populated.
- We never drop rows globally for missingness – we only filter per-analysis.

Recorded attacks by year



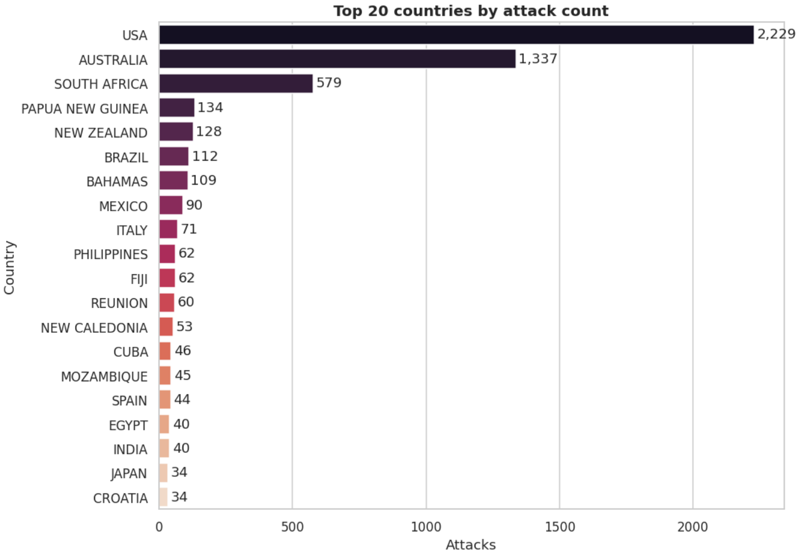
Almost all records are post-1900; reporting infrastructure improved dramatically through the 20th century.

Attack types

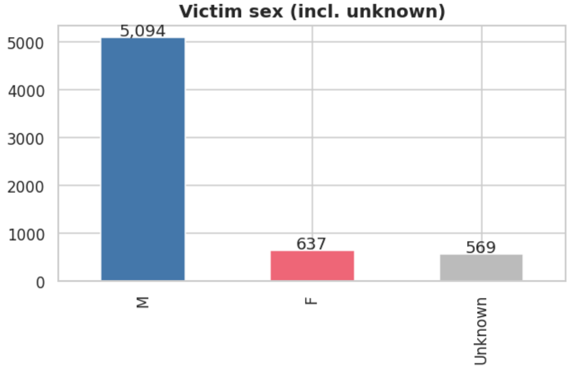


- **Unprovoked** dominates – the textbook "shark attack".
- **Provoked** cases (where the victim initiated contact) are a clear minority.
- **Sea Disaster** and **Boating** are rarer but include some of the highest-fatality events.

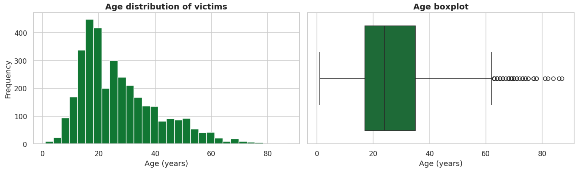
Where do attacks happen? (Top 20 countries)



Who are the victims?

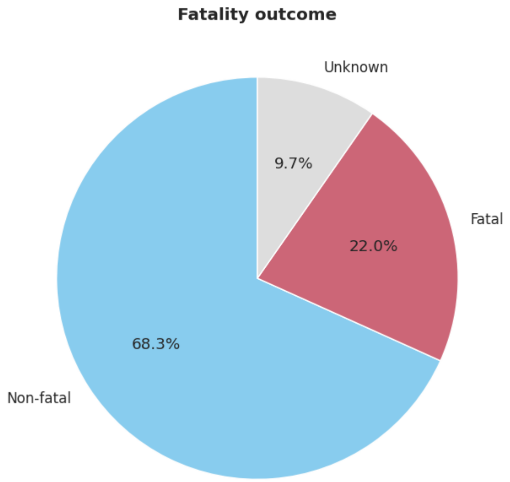


Massive male skew (∼8:1) when sex is known.



Median victim age ∼25; long right tail to elderly surfers / fishermen.

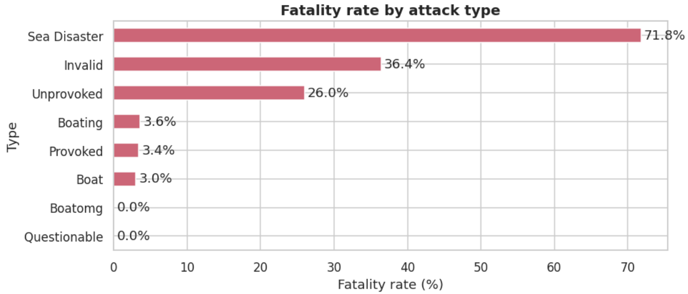
How often is an attack fatal?



Headline numbers

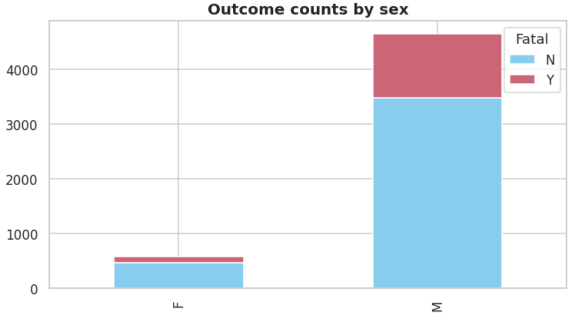
- Among **cases with a known outcome**, the long-run fatality rate is $\sim$**20%**.
- Roughly three quarters of survivable cases are recorded as non-fatal.
- Many old records have *no* outcome label – we exclude them from rate calculations.

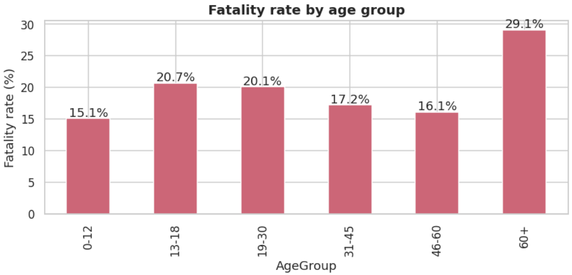
Fatality by attack type



- Sea Disasters are by far the deadliest – typically shipwrecks where many people are exposed for hours.
- Unprovoked encounters carry a moderate ~20 % fatality rate.
- Provoked cases are the least lethal – usually quick, defensive nips.

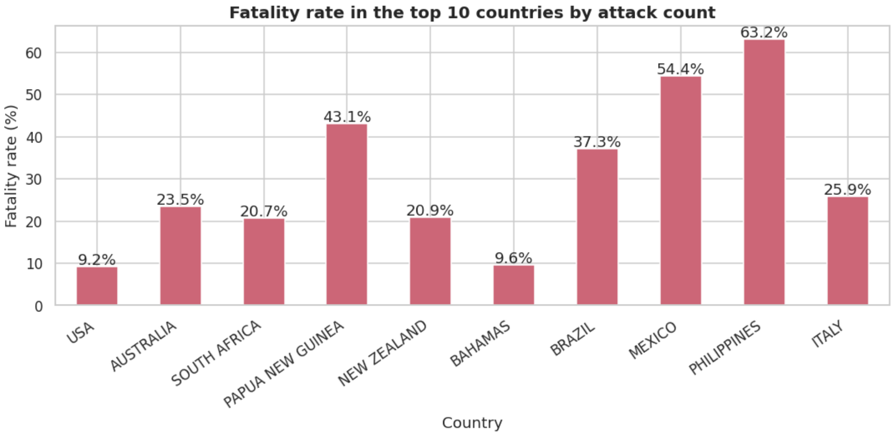
Fatality by sex and by age group





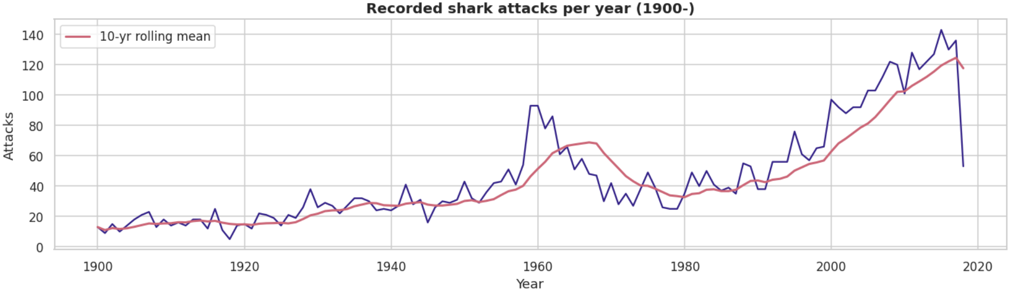
Fatality rate climbs with age – victims over 60 die at roughly twice the rate of teenagers, consistent with reduced ability to escape and recover.

Fatality in the top 10 countries



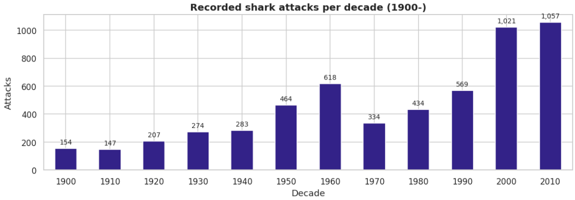
The USA records the most attacks but one of the *lowest* fatality rates – the inverse of countries with sparser medical access.

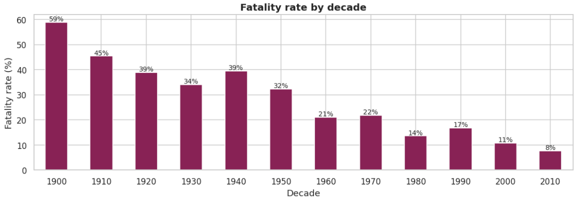
Recorded attacks per year, 1900–2018



Rolling mean shows a steady climb that accelerates after 1950 – mass tourism, surfing culture, internet reporting.

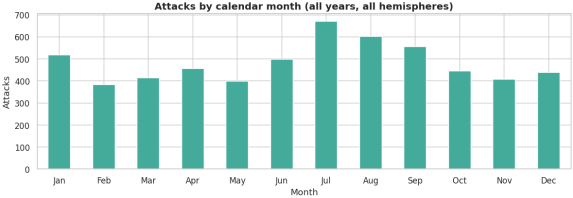
Attacks per decade and fatality rate over time



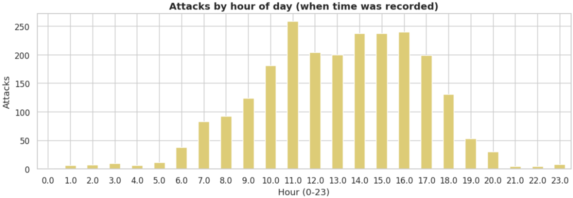


Attacks have multiplied; fatalities have collapsed. Pre-1950 fatality rates frequently exceeded 50 %; today they sit around 15–25 % thanks to faster rescue and trauma care.

When in the year? When in the day?

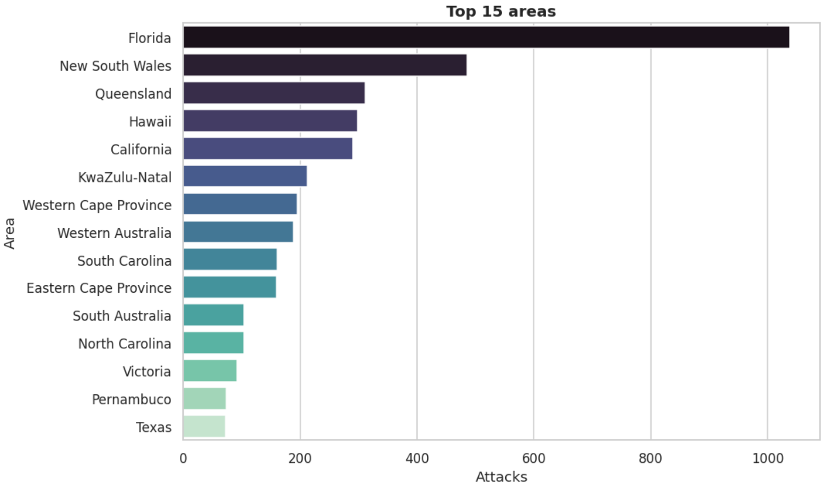


Peak in **July–August** – driven by the USA's heavy weight in the dataset.

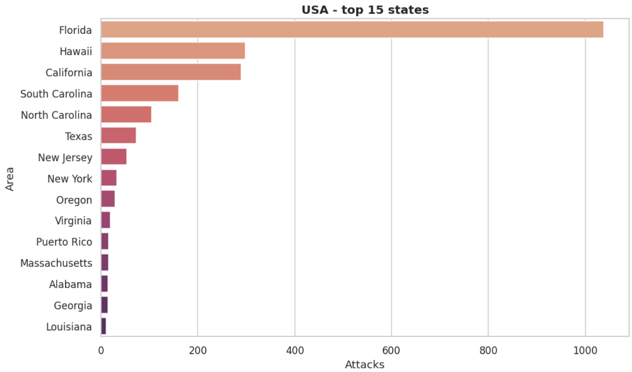


Bimodal: late morning (~11h) and late afternoon (~15–16h) – prime beach hours.

Top 15 areas worldwide

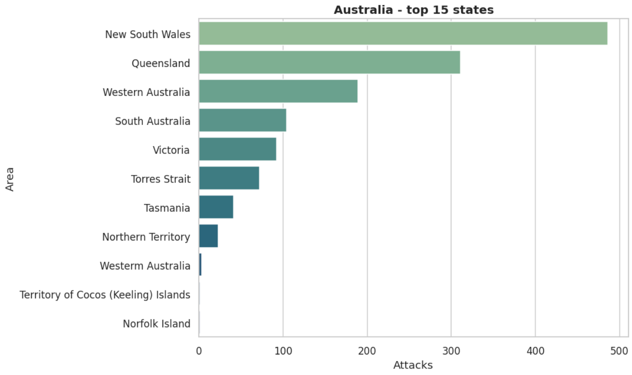


USA – top 15 states / regions



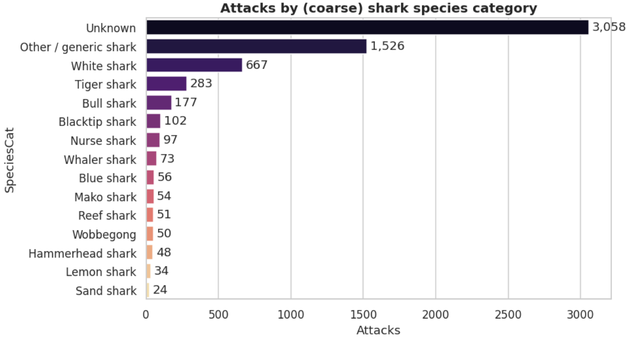
- **Florida is the global epicentre –** warm water, long coast, year-round beach activity.
- Hawaii, California and the Carolinas follow at much lower volumes.
- Florida alone accounts for more attacks than most countries on Earth.

Australia – top 15 states



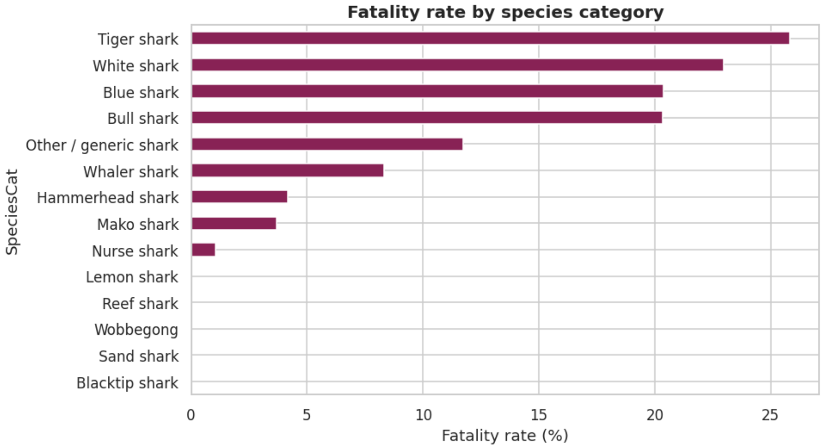
- **New South Wales and Queensland** dominate – Sydney/Gold Coast beaches.
- Western Australia has fewer attacks but a higher share of fatalities (white-shark waters).

Which species are involved?



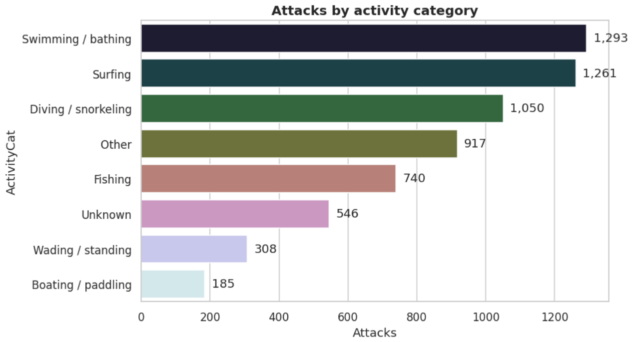
- Most cases either have no species recorded or only say *"shark"*.
- Among identified species, the "big three" – White, Tiger, Bull – account for the bulk of attacks.

Fatality rate by species



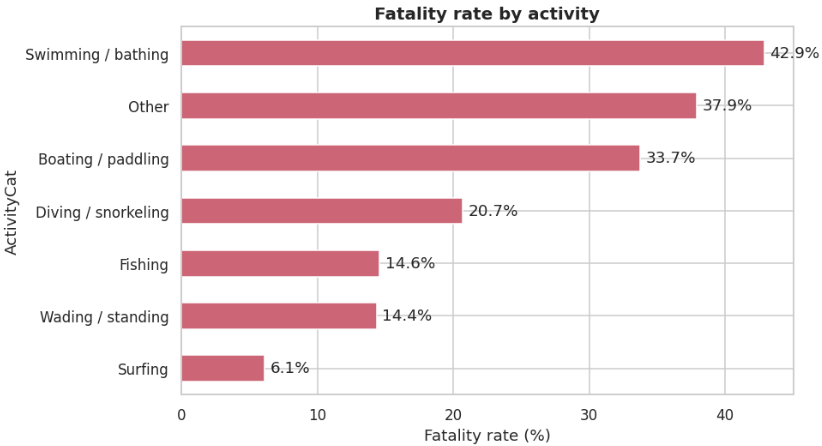
Tiger (∼26 %) and White (∼23 %) sharks top the lethality ranking among well-sampled species.

What were the victims doing?

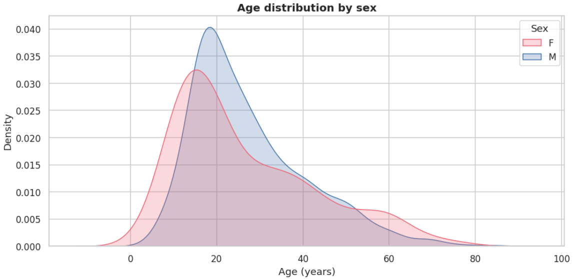


- **Surfing**, **swimming** and **diving/snorkeling** make up the majority – the activities that put people in shark habitat at vulnerable depths.
- Fishing/spearfishing carry above-average lethality (blood in the water).

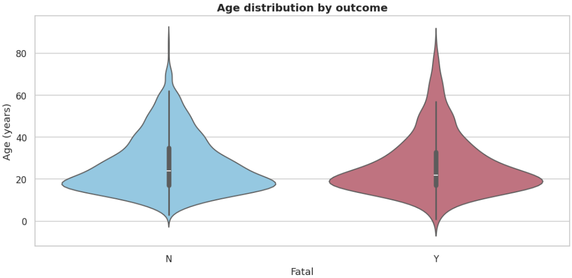
Fatality by activity



Age distribution by sex and outcome

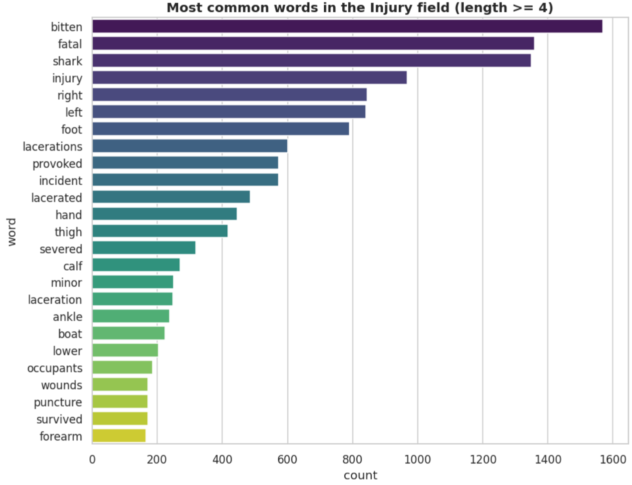


Both sexes peak in the late teens / 20s; men have a wider tail.



Fatal cases skew slightly older than non-fatal ones.

What words show up in the injury descriptions?



- Dominant tokens: **leg, foot, lacerations, hand, thigh, arm**.
- Confirms what the activity data implies – limbs in the water are bitten.
- Many cases note *"no injury"* – nudges and test-bites.

EDA – key insights

What the numbers say

1. **Reporting, not behaviour, drives the rise.** Records explode after 1950, but per-attack lethality *falls* steadily – the data reflects better media and medicine.
2. **Geography is concentrated.** USA + Australia + South Africa cover the majority of all incidents; **Florida alone** rivals entire countries.
3. **Victims are young men in the water.** ~87 % male, median age ~25, surfing/swimming/diving.
4. **Big three sharks dominate the lethal cases.** Tiger, White and Bull sharks carry the highest fatality rates.
5. **Sea disasters are an order of magnitude deadlier** than typical beach encounters.

Part II

Interactive Power BI Dashboard

Turning the EDA into a live, filterable story

From notebook to dashboard

Goals

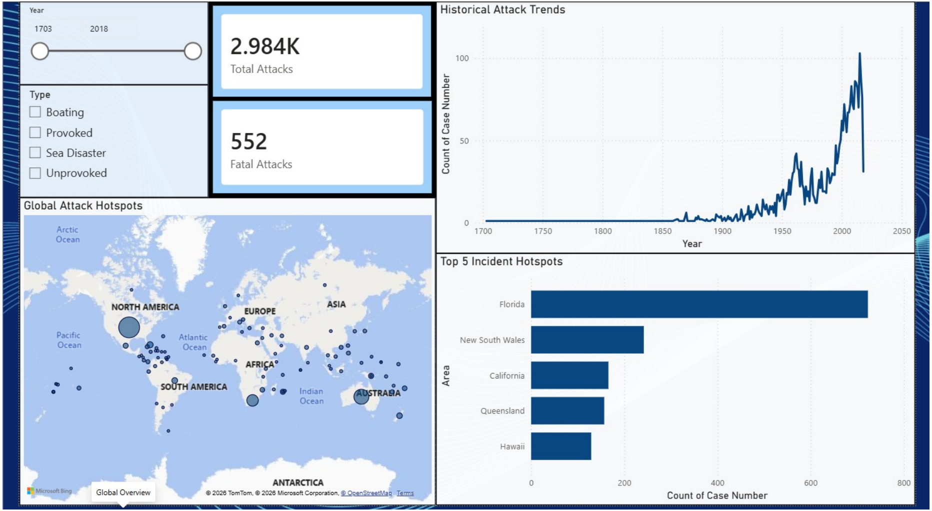
- Re-tell the EDA's story to a **non-technical viewer** in two screens.
- Let the user **filter freely** (year, country, attack type) and see every chart respond.
- Use a tight navy / sea-blue palette consistent with the dataset's nautical theme.

Structure

Page 1 – Global Overview When / where attacks happen, headline KPIs, hotspot map.

Page 2 – Victim Profile Who is attacked, doing what, and how often it is lethal.

Page 1 – Global Overview



Page 1 – the KPI cards

2.984K	**552**
Total Attacks	Fatal Attacks

- Headline counts pulled from the cleaned model; they re-compute live when the user moves the year slider or toggles attack types.
- Fatality ratio: $552/2{,}984 \approx \mathbf{18.5\%}$ across all recorded outcomes – consistent with the long-run EDA estimate.
- Two slicers on the left (Year range **1703–2018** and `Type`) drive every visual on the page.

Page 1 – historical trend

What we see

- Near-zero baseline until $\sim$1850.
- Slow growth 1900–1950, then a sharp climb peaking around 2015 ($>$100 attacks/year).
- Sharp drop at the right edge is the dataset cutoff (2018, partial year).

Why it matters

Same story as the notebook: **the curve mostly tracks reporting effort**, not shark behaviour.

Page 1 – where they happen

Global Attack Hotspots map

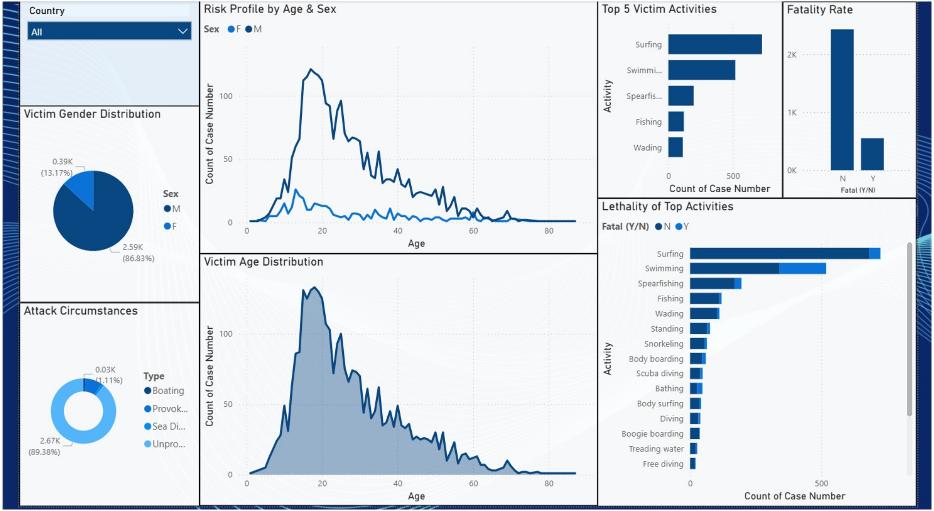
Bubble map – bubble size = count of attacks at that location.

- Three dominant clusters: **east & west USA**, **eastern Australia**, **South Africa**.
- Secondary clusters in Brazil, Reunion, Papua New Guinea.

Top 5 Incident Hotspots

1. **Florida** (∼700)
2. New South Wales (∼240)
3. California (∼180)
4. Queensland (∼170)
5. Hawaii (∼140)

Page 2 – Victim Profile



Page 2 – who is attacked?

Victim Gender Distribution

- **Male: 86.83%** (2.59K cases)
- **Female: 13.17%** (0.39K cases)

A ∼**6.6:1 male-to-female ratio** – reflects exposure (surfing, fishing, diving demographics) more than any biological preference of the sharks.

Attack Circumstances

- **Unprovoked: 89.38%** – the classic case.
- Sea Disaster + Provoked + Boating ≈ 10 %.

Donut format reinforces how rare "the shark started it" really is in relative terms.

Page 2 – the age story

Risk Profile by Age & Sex

- Both curves peak sharply in the **mid-teens to mid-twenties**.
- Male curve dwarfs the female curve at every age bin – but the *shape* is similar.
- Long tail to age 80+, mostly male surfers and fishermen.

Victim Age Distribution

- Right-skewed distribution centred around age $\sim$18–22.
- Median age ≈ 25, matching the notebook.
- Children ≤ 12 are rare – close adult supervision in shallow water.

Page 2 – activity and lethality

Top 5 Victim Activities

1. **Surfing** (largest)
2. **Swimming**
3. Spearfishing
4. Fishing
5. Wading

Lethality of Top Activities (stacked Y/N)

- Surfing has the *highest absolute count* of attacks but a *moderate* share of fatalities – short waters, fast rescue.
- **Swimming** and **spearfishing** carry a noticeably higher fatal share.
- **Boating**-related and **free-diving** categories are low-count but lean fatal.

Page 2 – the overall fatality split

Fatality Rate card

A simple **Y vs N** bar that re-computes under any filter combination.

- Default (all years, all countries, all types) shows the dominant **N** (non-fatal) bar.
- Filtering to `Type = Sea Disaster` flips the ratio: fatalities catch up to non-fatalities.

The interactive power

Because every visual on the page is wired to the same filters, the user can ask, in one click: *"What does an attack on a swimmer in South Africa look like, and how often is it fatal?"*

Dashboard takeaways

What the Power BI view adds to the notebook

- **Two screens, two questions** – "where/when" then "who/how" – replace 13 notebook sections with a clean narrative.
- **Slicers turn statements into questions.** A reader can re-derive any of our EDA findings live: filter to `USA`, see Florida dominate; filter to 1900–1950, watch the fatality split flip.
- **KPIs anchor the eye.** The two big cards (2,984 attacks, 552 fatal) give a viewer their bearings in under a second.
- **Consistent colour grammar** – blue = safe / non-fatal, light blue accent = subgroup, dark navy = headline KPI.

Overall conclusions

1. Shark attacks remain **rare** and, conditional on happening, **less and less lethal** over time.
2. Risk is overwhelmingly concentrated in a small number of **warm, populated coastlines**.
3. The "typical" victim profile (**young adult male, surfing or swimming, in the afternoon**) is so dominant that prevention messaging can target it very precisely.
4. The dataset is a strong showcase for the visualisation pipeline: *clean* $\rightarrow$ *explore in Python* $\rightarrow$ *tell the story in Power BI*. Both layers are necessary – the notebook for rigour, the dashboard for reach.

Thank you!

Questions?